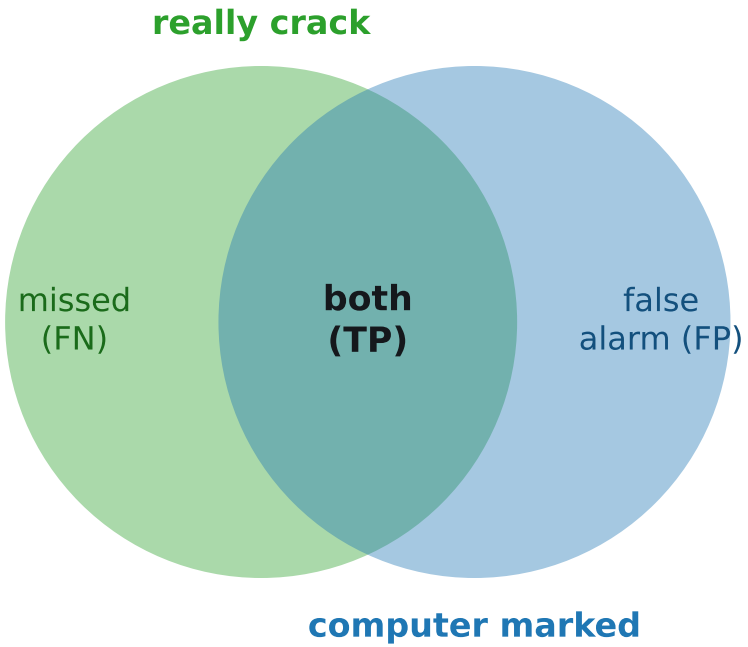
Crack Fractography — where the measurements stand

- v1.49.2. Every figure here is generated from a committed record by a script in this repository; none is drawn by hand and none carries a number that was not measured.

⚠ Eight slides. Read each one with its caveat: most of these results were over-read at some point for exactly the reason stated underneath.

All four scores are fractions of the same three areas

Green is the crack you painted. Blue is what the computer marked. Everything else is just which part of this picture you divide by.



Precision

both / everything BLUE

did it cry wolf?

Recall

both / everything GREEN

did it miss things?

F1

balances the two above

one number, not cheatable

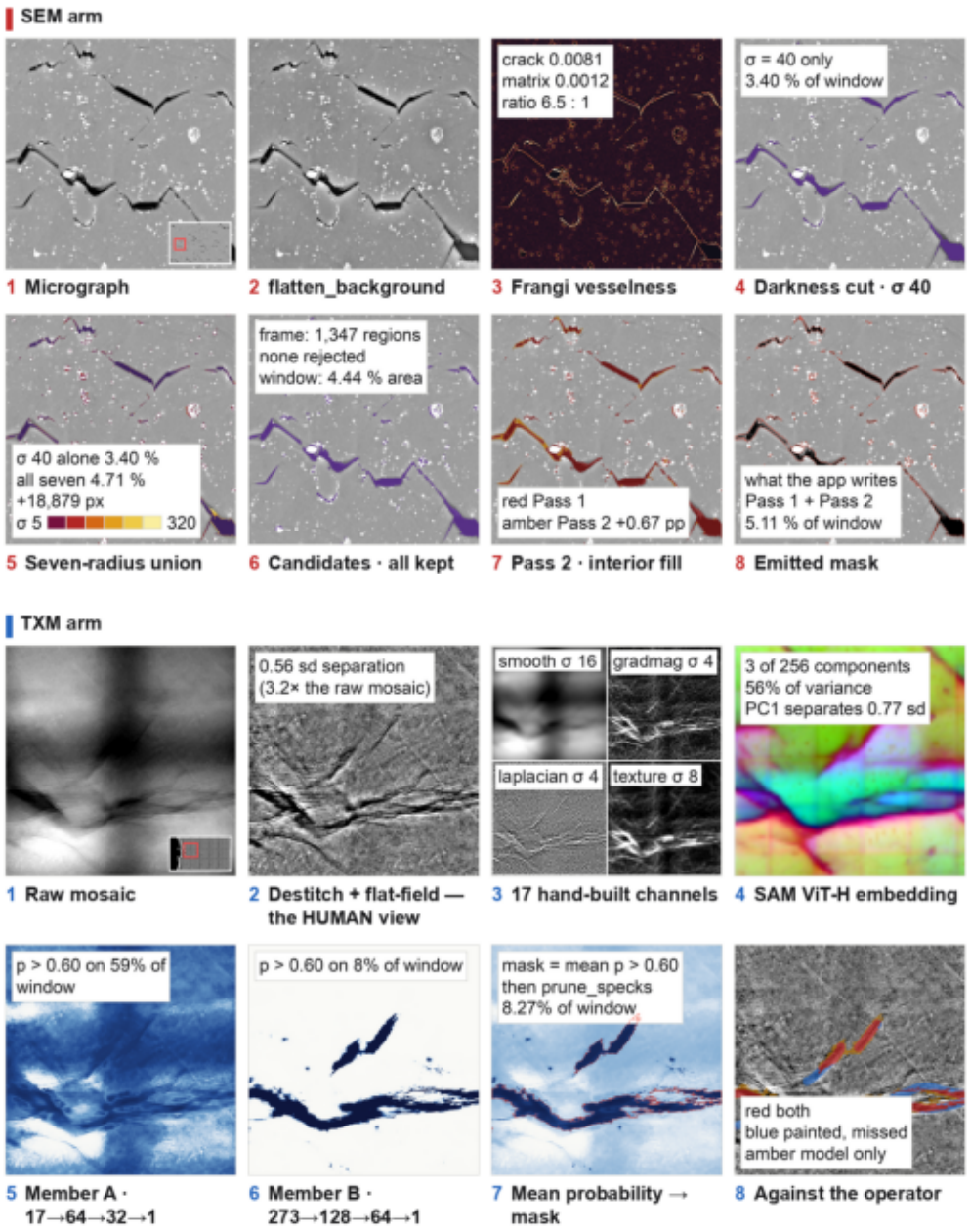
IoU

both / the WHOLE picture above

strictest; rewards thickness

The words, in plain English

Precision	Of everything the computer marked as crack, how much really was crack. High precision with low recall just means it only marked the easy bits.
Recall	Of all the crack that is really there, how much the computer found. MARKING THE WHOLE IMAGE SCORES A PERFECT 1.0. Recall alone is never a score.
F1	The two above combined into one number, in a way that punishes being bad at either. You cannot cheat it by marking everything -- that destroys precision, so F1 collapses.
IoU (intersection over union)	The overlap divided by everything that either of you marked. The strictest of the four. It rewards drawing THICK. A perfectly placed thin line scores badly against a thick label, so a fatter, less accurate answer can beat a correct one.
clDice	Did you find the crack along its LENGTH -- following the path, ignoring how thick you drew it. On these labels it is still partly cheatable by thickening: a no-information dilation moved it +0.048 on 13 of 13 specimens.
Skeleton-F1 at 2 px	The same idea as clDice but done properly: compare the two centre-lines, allowing a 2 pixel wobble. The one metric here that a thicker mask does NOT improve (+0.0015, p=0.68).
Tolerance	How many pixels of slack before a pixel counts as wrong. '2 px' forgives being two pixels off. It changes the answer enormously, so a score without a stated tolerance is unreadable.
The floor (no-information level)	What a method that knows nothing would score, by marking at random. A precision of 0.26 sounds poor; if the floor is 0.08 it is 3x better than guessing.



What each stage does

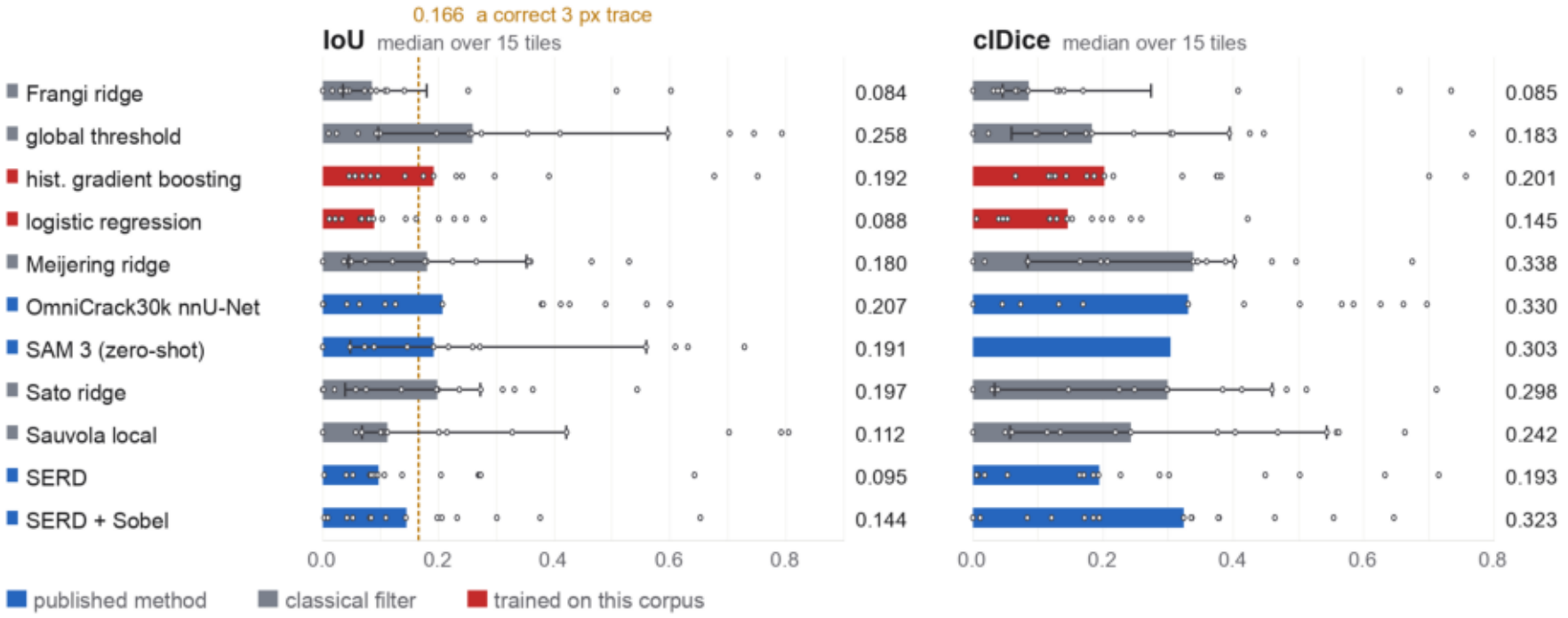
- The pipeline the app runs, panel by panel.
- Seven flattening radii, unioned.
- Every candidate region kept.

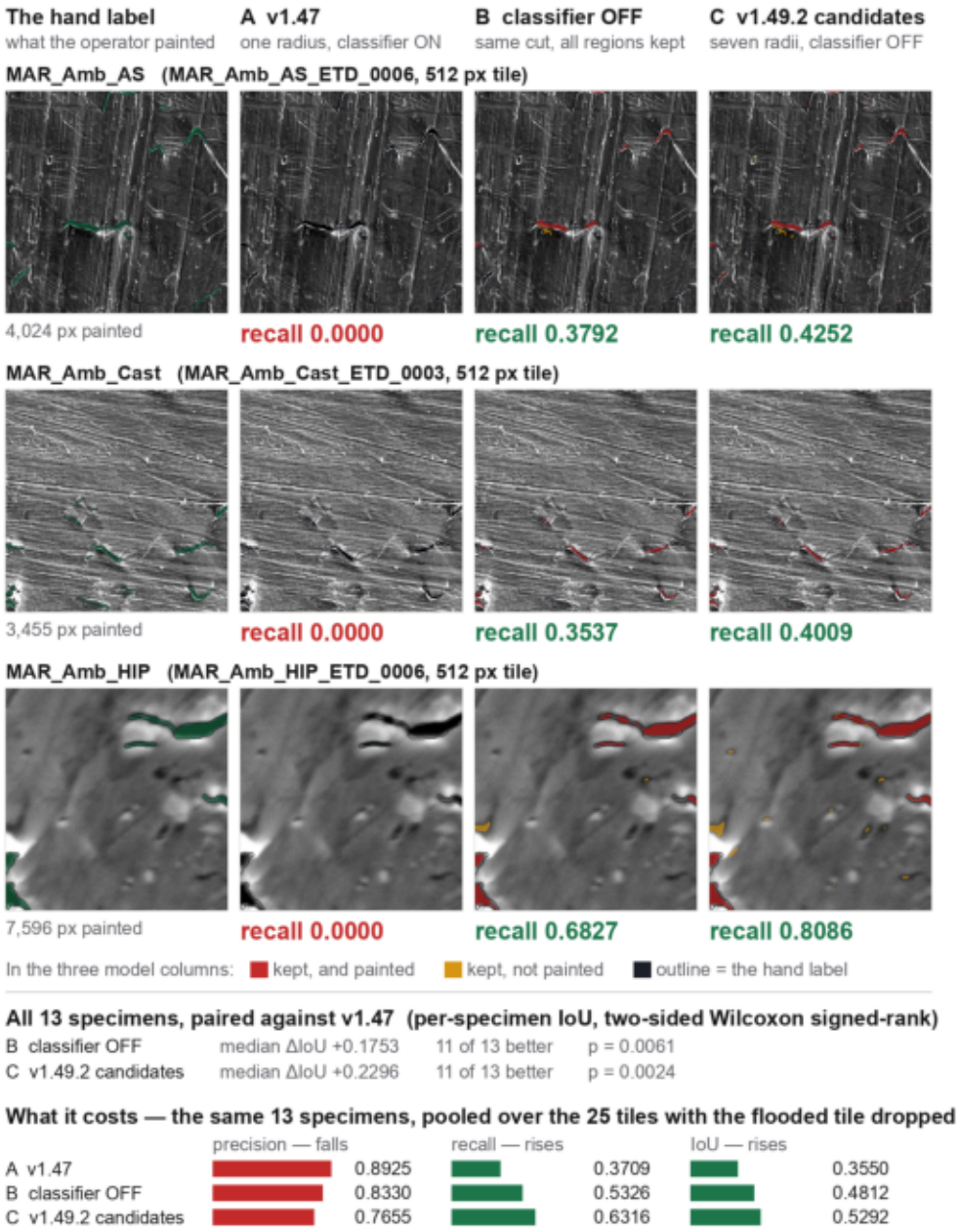
⚠ The retired classifier would have deleted 21.6% of this.

Every method, identical data

- 11 arms, same 15 tiles, same budget.
- The ranking REVERSES between the two metrics.

⚠ Dashed line = a correct 3 px trace. Above it means fatter, not better.





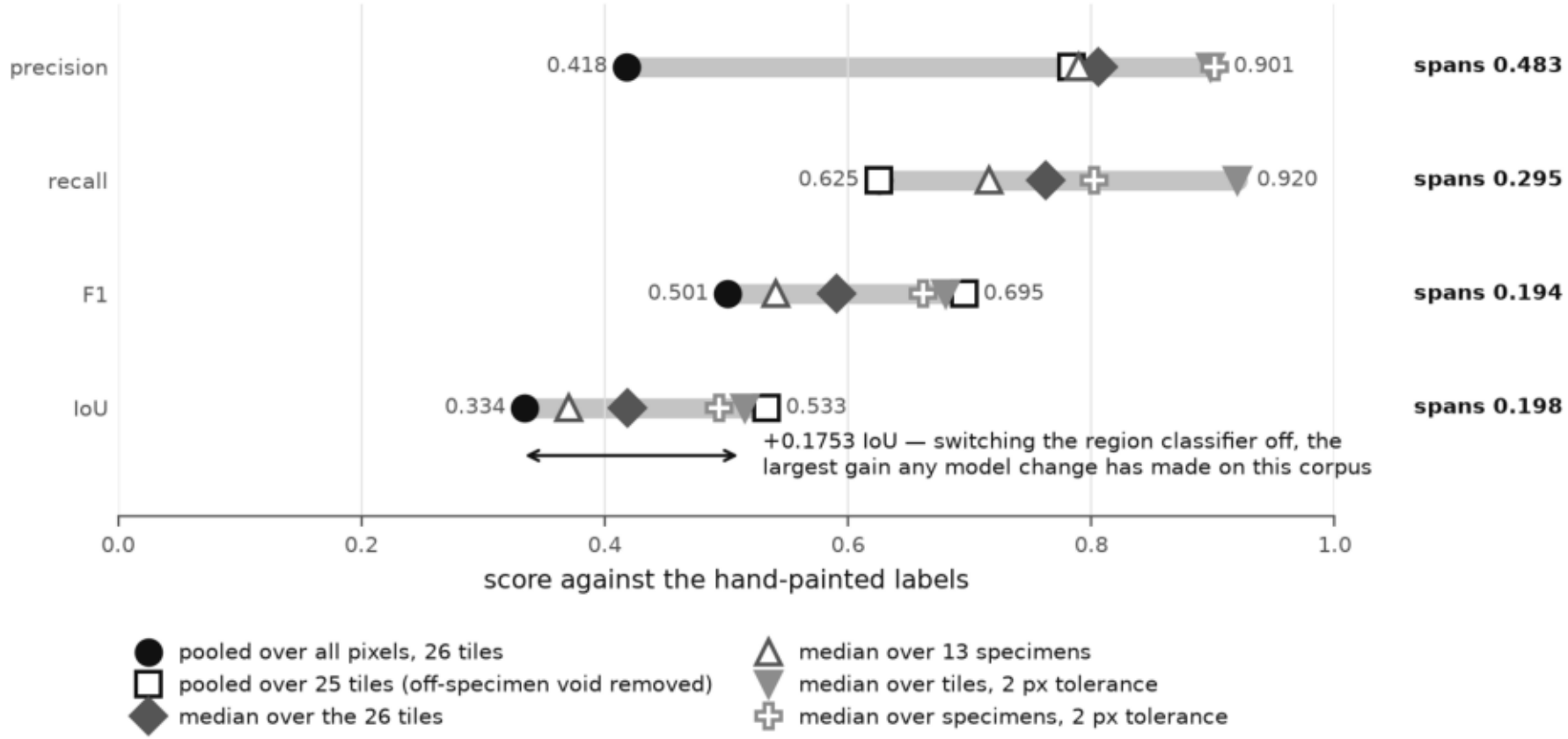
What deleting the classifier recovers

- Three specimens went from recall 0.000 to 0.35-0.68.
- The cracks were always found; the model rejected them.

⚠ It costs precision on clean fields: 0.89 to 0.78.

One set of masks, six ways of reporting it

v1.49.1, region classifier off, 7-scale union, mad_k 5.0, absolute_dark 10. Nothing about the detector changes between the six readings — only the rule for turning per-tile counts into one number.



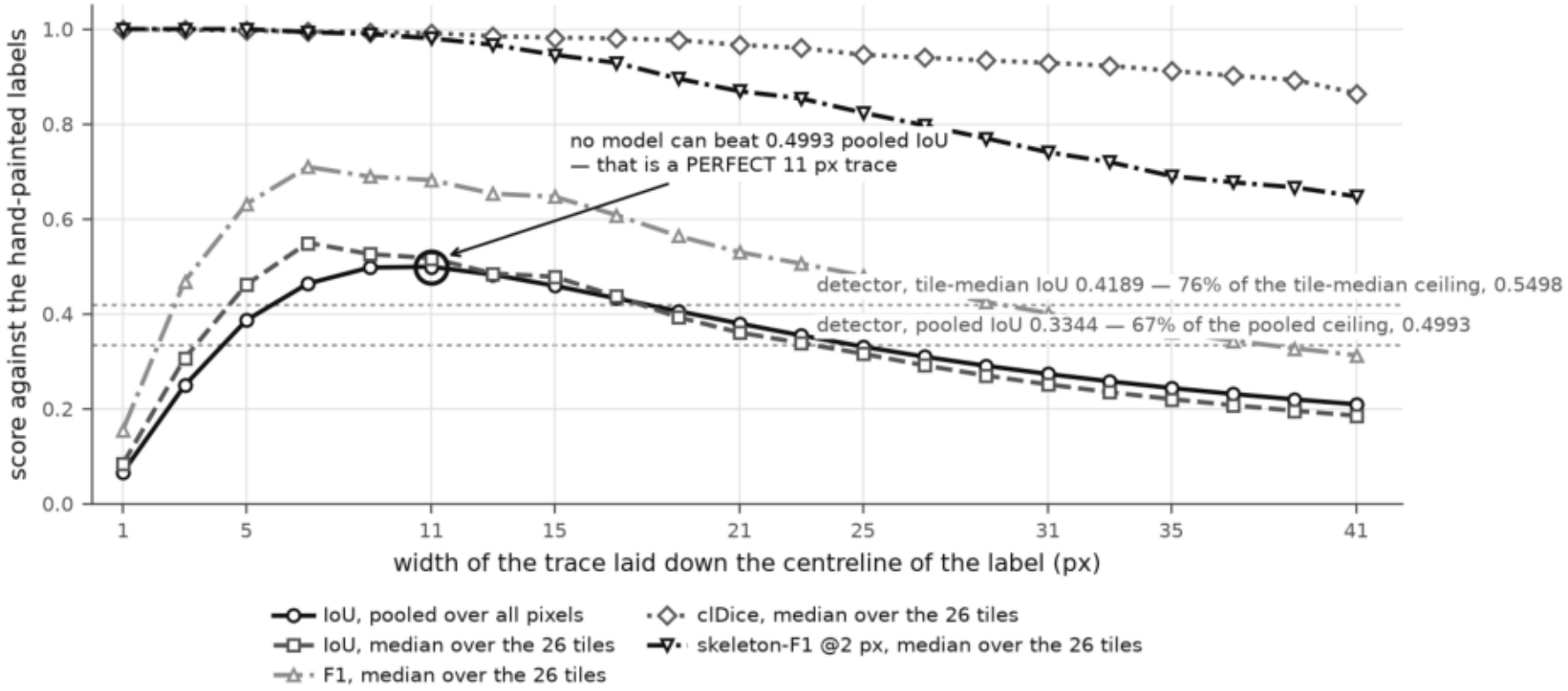
How you report it matters more than the method

- One set of masks, six ways of reporting it.
- IoU spans 0.1985 across them.

⚠ That span is larger than any method change measured here.

What a PERFECT answer scores, and why IoU cannot reach 0.8 here

Every mark is the human's own paint, skeletonised and redrawn at a chosen width — the right answer, if the paint says where the crack is but not how wide the brush was. The overlap metrics still score it far below 1, and the ceiling moves with the reporting rule.



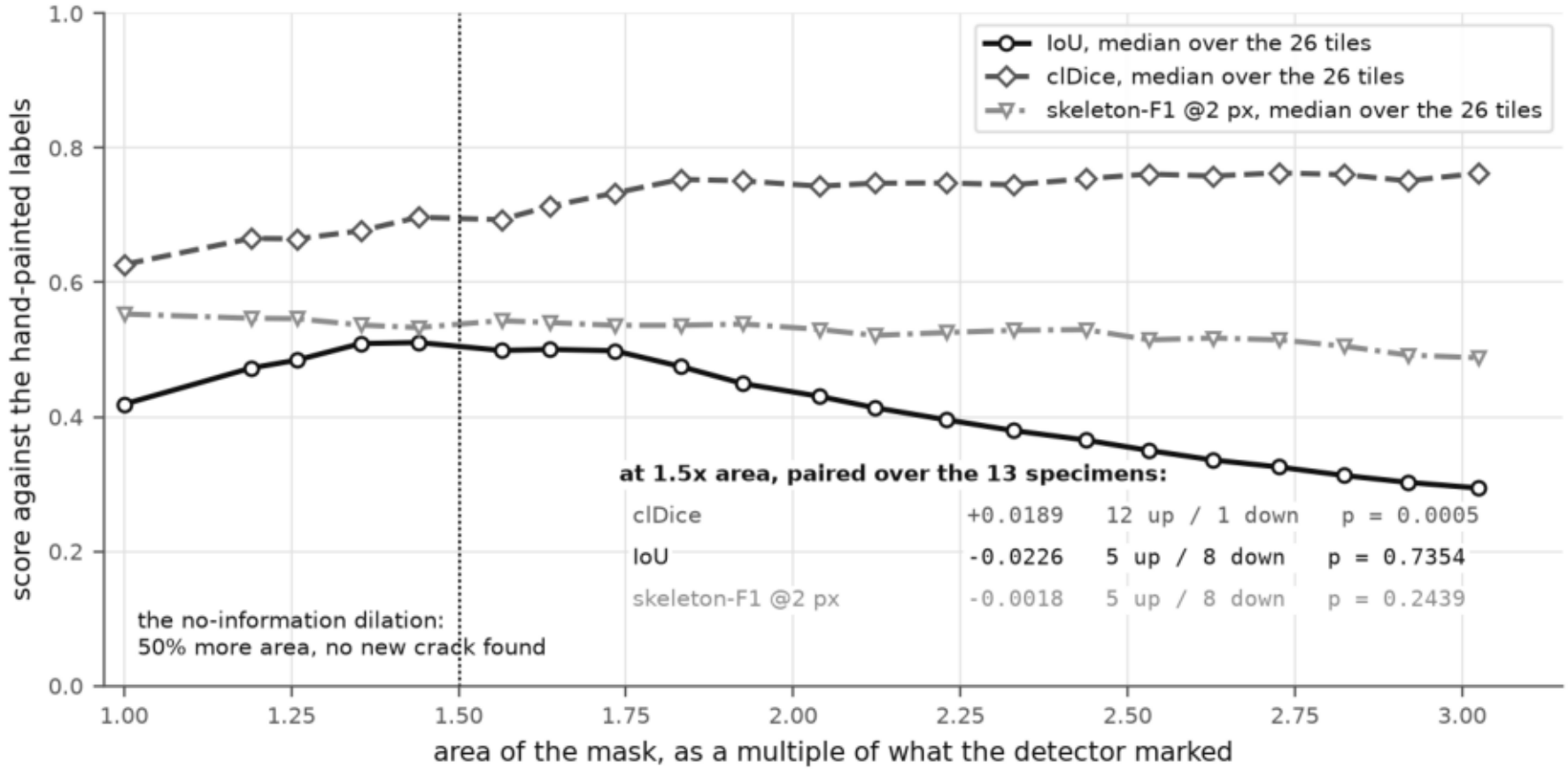
A correct answer caps at IoU 0.50

- A perfectly placed trace peaks at 0.4993.
- So IoU 0.42 is ~75% of possible, not 42%.

⚠ Always quote IoU with this ceiling.

Which metrics can be bought by simply drawing fatter

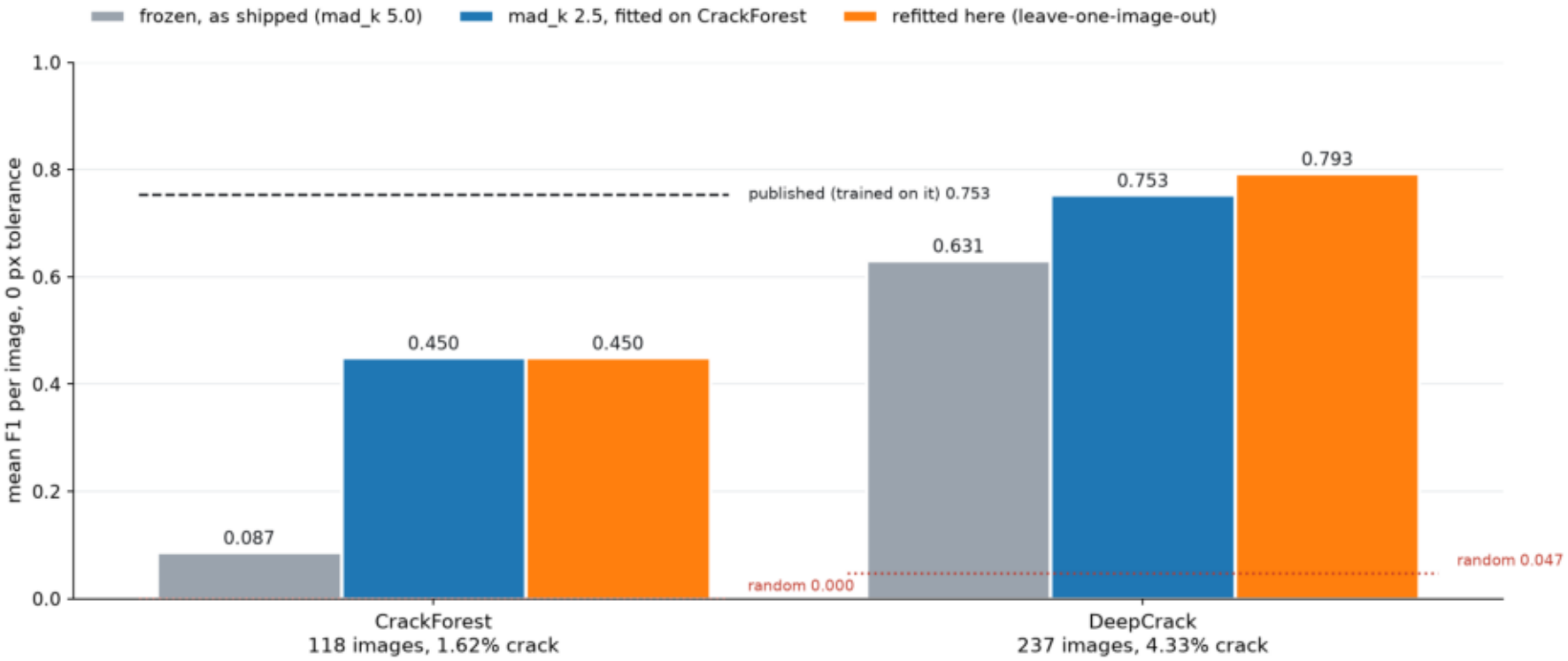
The shipped mask, grown outward at a continuous radius until it holds the stated multiple of its own area. Growth adds no information: it cannot find a crack the detector missed. A metric that RISES under it is measuring thickness, not placement.



The metrics reward thickness

- Dilating the mask adds no information.
- It still moves clDice +0.048 on 13 of 13.
- Skeleton-F1 moves +0.0015.

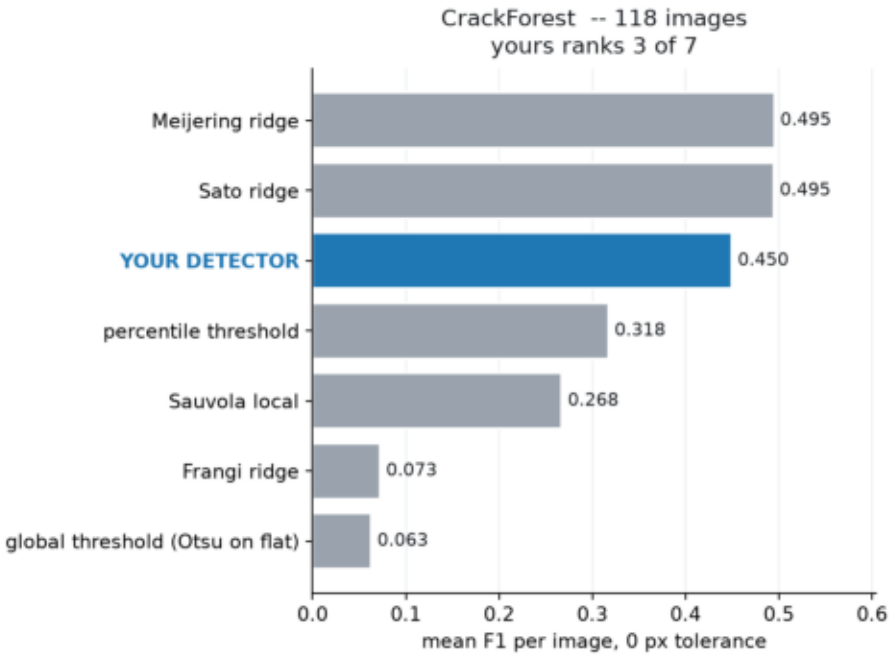
⚠ IoU, clDice and the paint gate all have this fault.

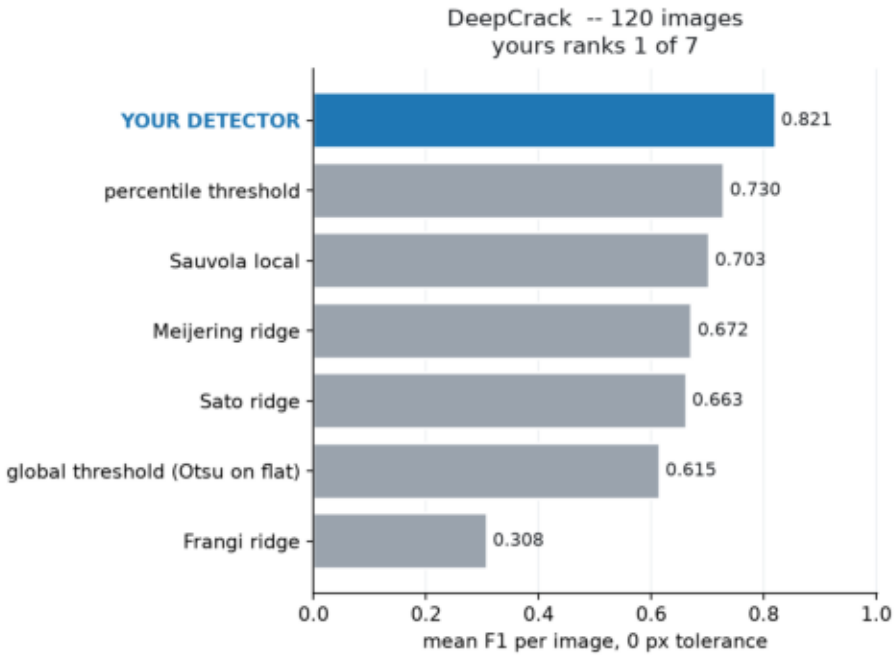


On public datasets

- Frozen: 0.631 on DeepCrack, 0.087 on CrackForest.
- Refitting one constant: 0.793 and 0.450.

⚠ CrackForest's fitted value reaches 95% of refitting on DeepCrack.





Against six classical baselines

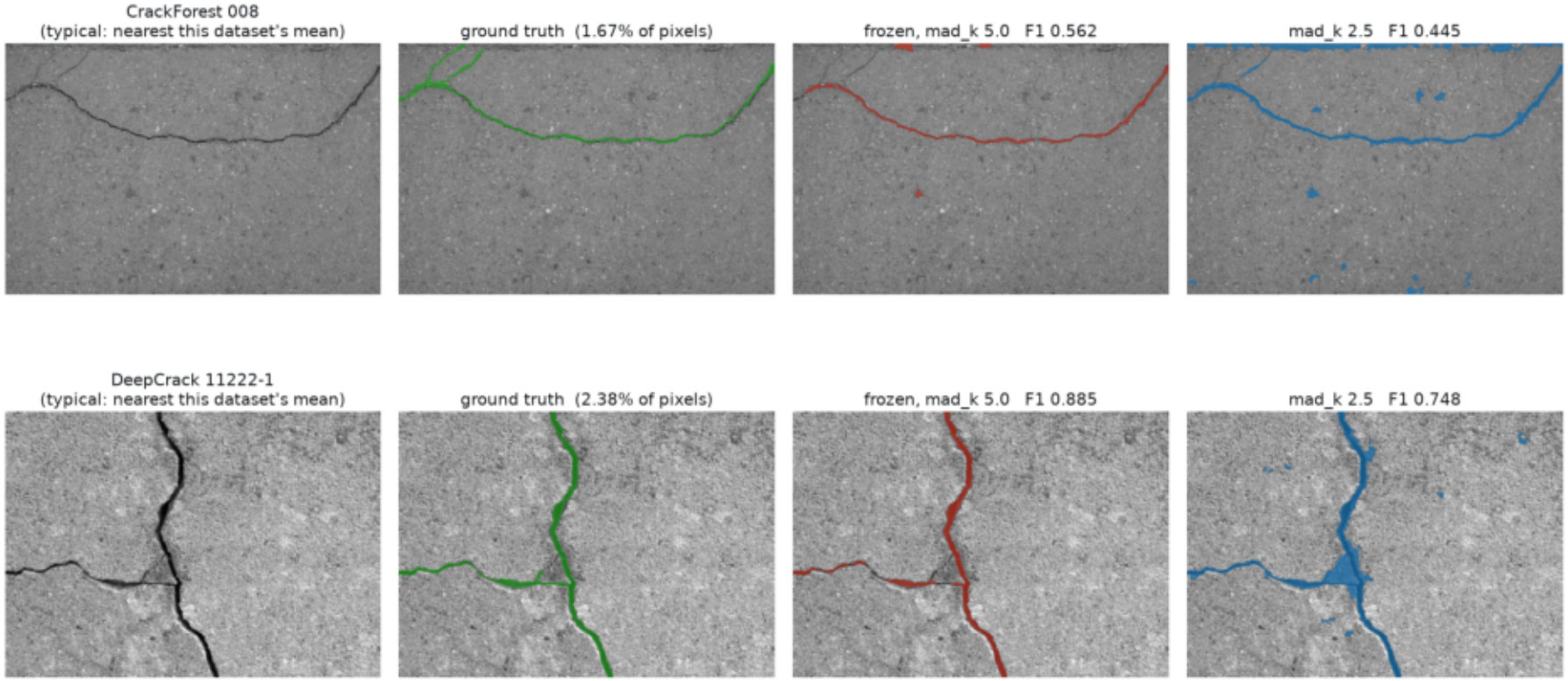
- 1st of 7 on DeepCrack (0.821 vs 0.730).
- 3rd of 7 on CrackForest, behind two ridge filters.

⚠ Better on wide bright cracks, worse on thin faint ones. Panels are 118 and 120 images -- don't read across.

The same, on the images

- Ground truth, frozen, and refitted.
- One typical image per dataset.

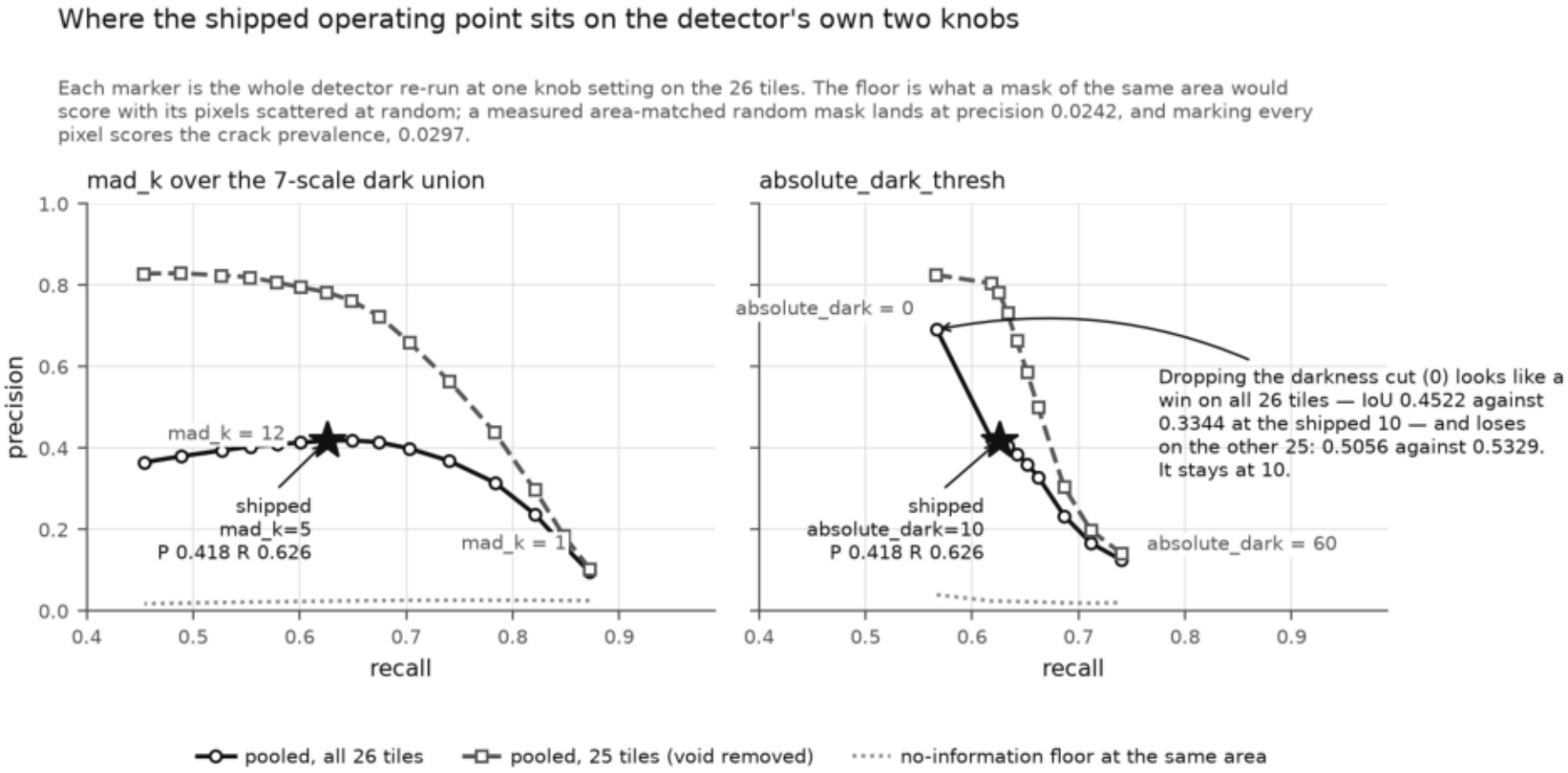
⚠ Images picked by rule (nearest the mean), not by eye.



Recall alone is not a score

- The loosest setting reaches recall 0.98.
- Its F1 is 0.162 and its precision equals plain Otsu.

⚠ Read precision against the floor: 0.081 on TXM, 0.030 on your tiles.



What each measurement means (1 of 3)

Read straight out of app/groupstats.py, so this slide cannot drift from what the application shows.

Crack count

cracks

How many separate crack regions were found. A COUNT, NOT AN AMOUNT: one long crack and a hundred specks both add to it. Read it beside 'Largest crack's share of area'.

Largest crack's share of area

fraction of crack area

Of all the crack area in a frame, the fraction sitting in the single biggest region. Near 1.0 means one dominant crack; near 0 means a genuinely distributed network. This is what tells you whether the crack count means what you think.

Number density (per megapixel)

cracks / Mpx

Cracks per million pixels of image. Comparable only between frames at the SAME magnification, because a pixel is a different physical size at each.

Number density (per mm2, scale needed)

cracks / mm2

Cracks per square millimetre of specimen surface. The physical version of the row above, so it IS comparable across magnifications -- but only frames with a recovered pixel size contribute.

What each measurement means (2 of 3)

Read straight out of app/groupstats.py, so this slide cannot drift from what the application shows.

Total crack area (pixels)

px2

Every crack pixel in the group added up. Summing this across frames of different magnification adds areas that are not the same physical size.

Total crack area (um2, scale needed)

um2

The same total converted to physical area. Only frames with a known pixel size contribute, so this can cover fewer frames than the pixel row above it.

Crack area fraction

% of frame

What percentage of the imaged area is crack. Dimensionless, so it is the one size measure that compares fairly across magnifications and modalities.

Average crack area

px2

Mean area of one crack region. Pulled upward by a single large crack; compare it with the median below.

What each measurement means (3 of 3)

Read straight out of app/groupstats.py, so this slide cannot drift from what the application shows.

Median crack area

px2

The typical crack's area -- half are smaller, half larger. Unlike the mean, one huge crack cannot move it.

Average length (all)

px

Mean crack length along its centreline, INCLUDING cracks that run out of the frame. Those are measured only as far as the frame edge, so this is a lower bound.

Average length (uncensored only)

px

The same average over only those cracks fully inside the frame, so no length is cut short. If this is much larger than the row above, the frame is clipping your cracks.

Average aspect ratio

length / width

How elongated a region is: length divided by width. Around 1 is round -- porosity rather than a crack. Painted crack in this corpus runs about 3.3; painted porosity about 1.5.

arm	IoU	clDice	basis
>>> shipped v1.49.2, all 1	**0.3628**	**0.4246**	**no parameter fitted here; config informed by 3 shared frames**
>>> shipped v1.49.2, fran	**0.3530**	**0.3658**	**n=9; frames never used to choose the config**
>>> shipped v1.49.2, spe	**0.1596**	**0.2883**	**n=2; too small to conclude, trends against**
global threshold	0.2579	0.1827	one line of code, LOFO
NESTED LOFO ensemble	0.2526	0.1739	honest nested selection
Sato ridge	0.1972	0.2981	classical filter, LOFO
SAM 3 (zero-shot union)	0.1914	0.3030	foundation model, no fitting
OmniCrack30k nnU-Net	0.2069	0.3304	trained on 30,017 images incl. steel
Meijering ridge	0.1802	0.3382	classical filter, LOFO; best clDice of the competitors
Sauvola local	0.1117	0.2421	classical filter, LOFO
Frangi ridge	0.0843	0.0849	classical filter, LOFO
a correct 3 px trace	0.1662	0.9997	the physically correct answer, not ceiling reference

How good is this model, against published ones?

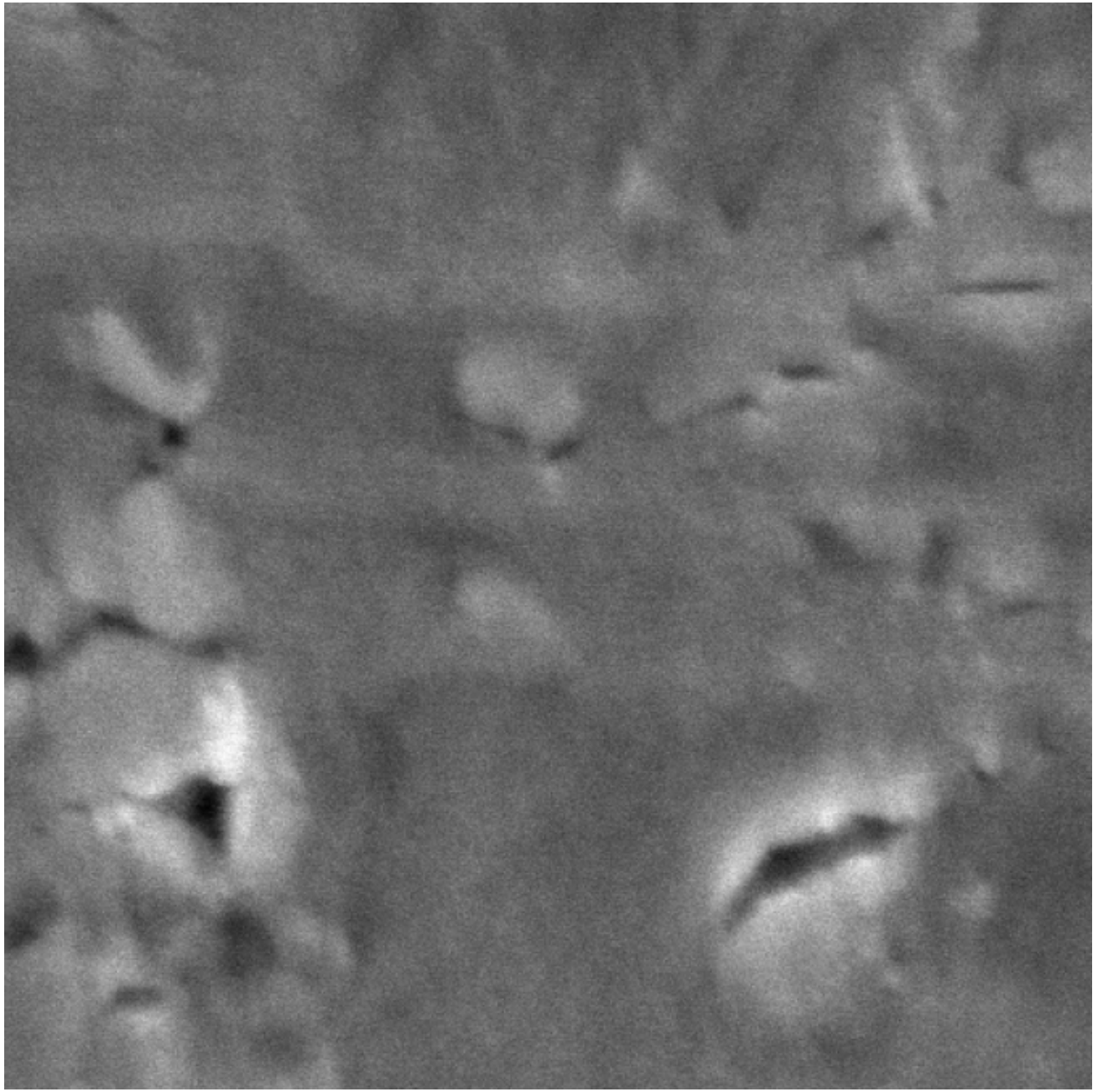
On the 15 tile benchmark the published models were scored on, the shipped detector leads both metrics. The lead survives dropping the frames whose tiles informed its configuration, and does not survive dropping the specimens - n=2 there, which settles nothing either way.

⚠ Every competing arm is scored leave-one-FRAME-out, which penalises it for having a learnable parameter. v1.49.2 has no parameter fitted on these tiles, but 3 of the 9 frames and 5 of 7 specimens also appear in the 26-tile set used to choose its configuration, so the all-15 row is the favourable reading. And at n=15 nothing on this benchmark separated statistically in the first place.

The pipeline, one real image per stage

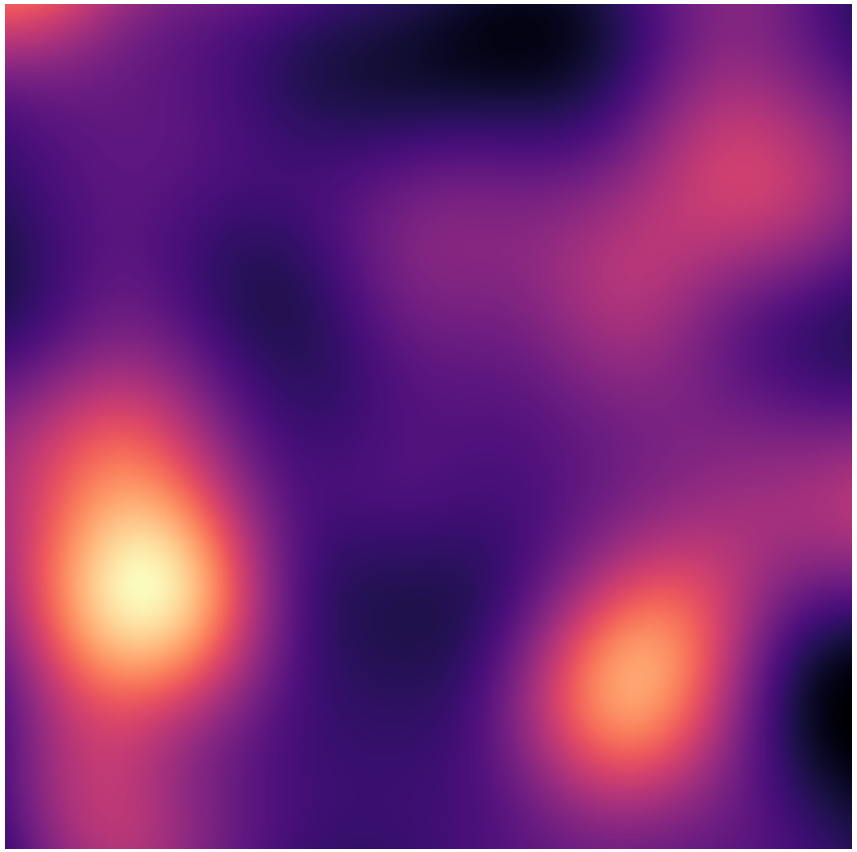
- Frame HIP_24hr_SE_Side_006, window 262,144 px. Every panel is a real array from the real pipeline -- no mock-ups, no schematic boxes.

⚠ The frame is chosen by a rule that works against the detector: the lower of the two central tiles by IoU, tie broken downward, so the pages cannot flatter it.

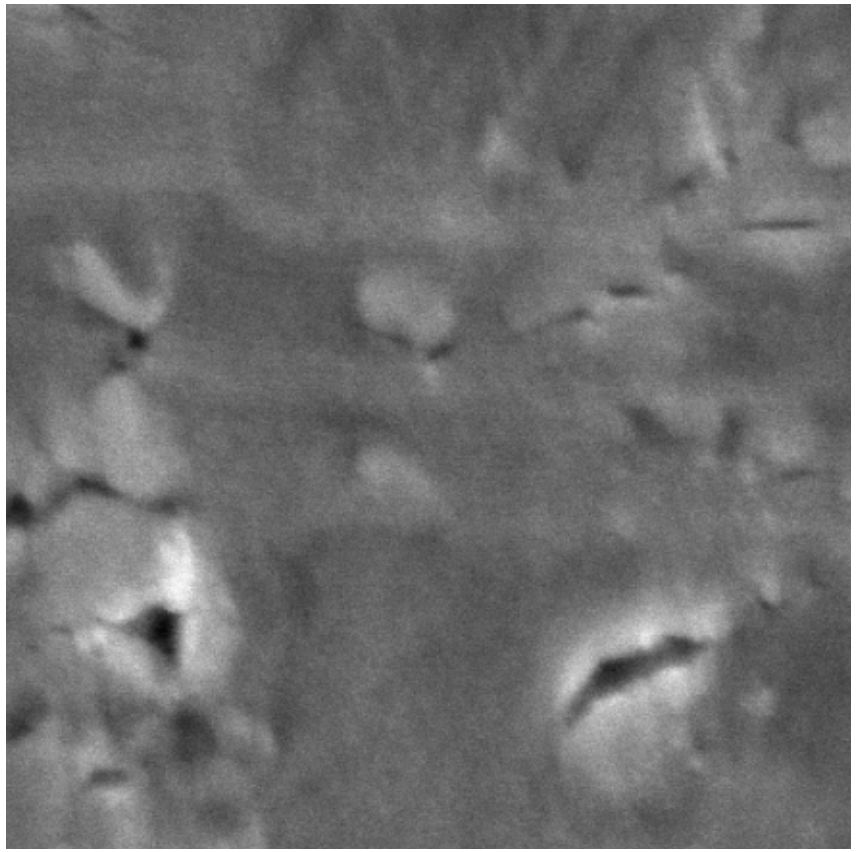


1. The micrograph

- What it does. Reads the 16-bit TIFF, stretches it to 8 bit on its 0.5/99.5 percentiles, and crops the databar off: 6144×4096 px survive.
- Cost. A percentile stretch, not min/max: a hot pixel or the crack itself at 0 would otherwise own the range and flatten the background texture Otsu...



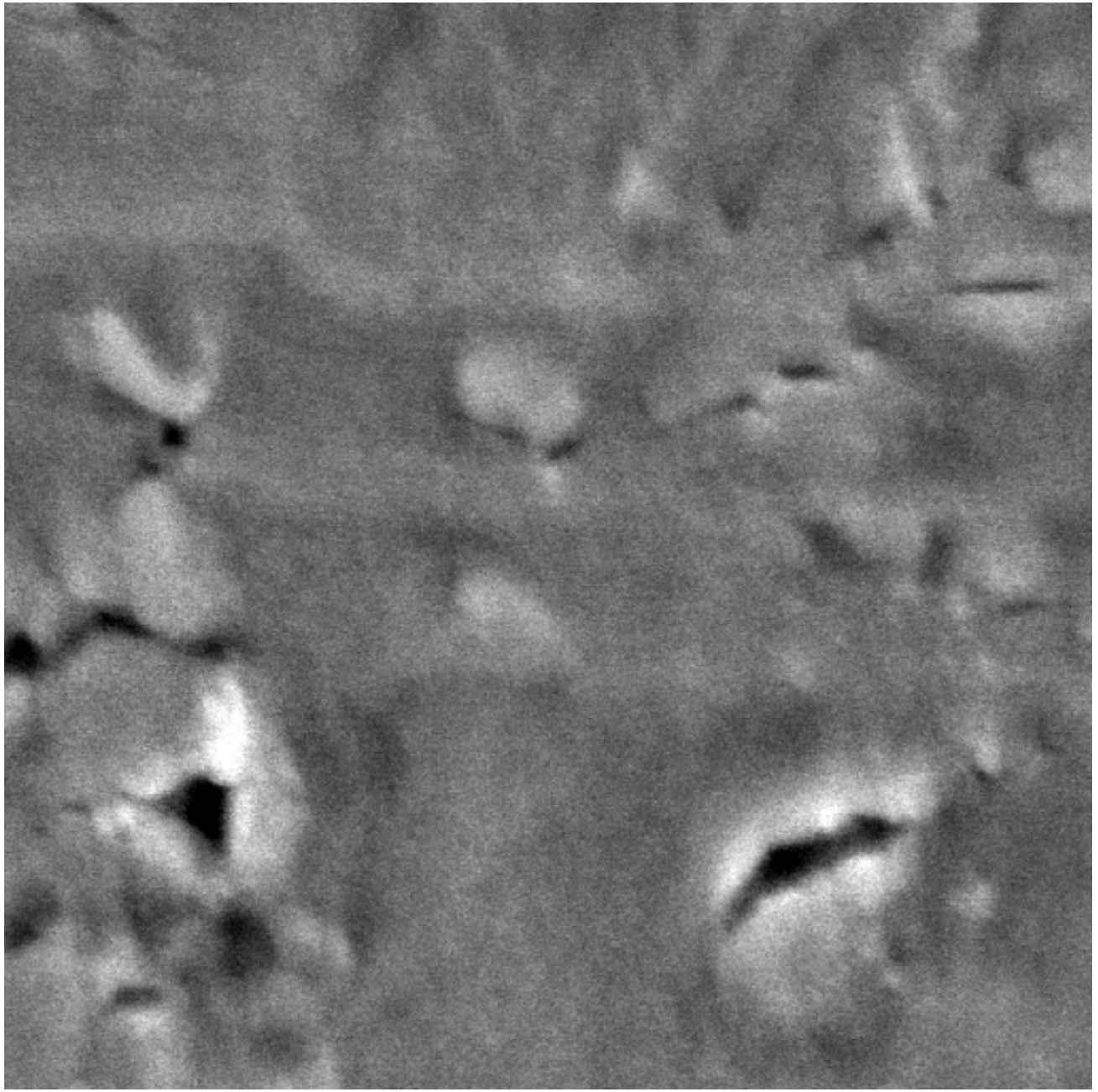
The Gaussian background estimate at sigma=40 over this crop, false-coloured. It spans 92.8 to 139.3



The same crop, as captured, for comparison.

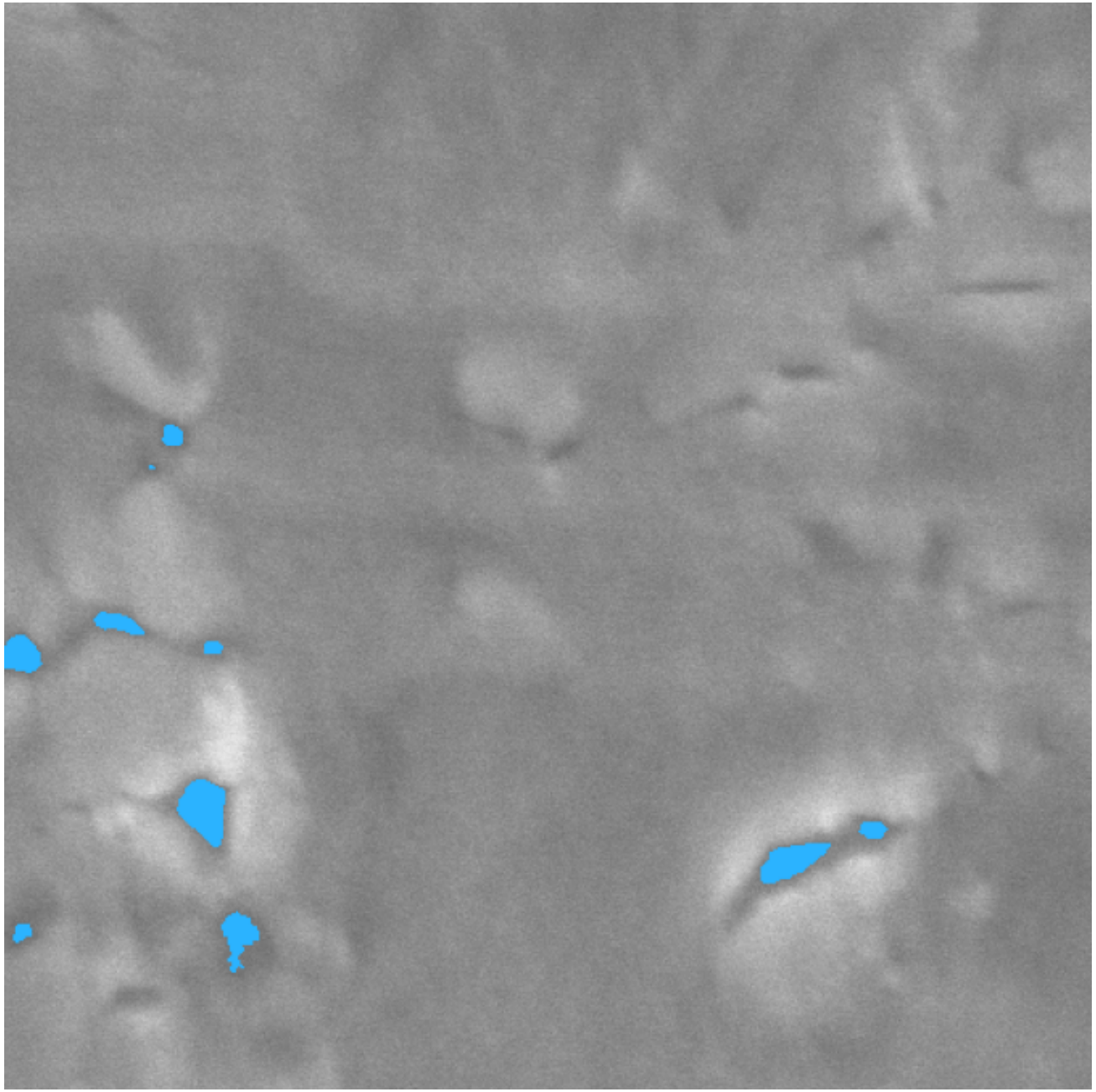
2. Why the next step exists: the illumination

- What it does. Nothing yet — this is the quantity flatten_background is about to subtract.
- Why it matters. A single global threshold on the raw image would track this drift instead of the crack.



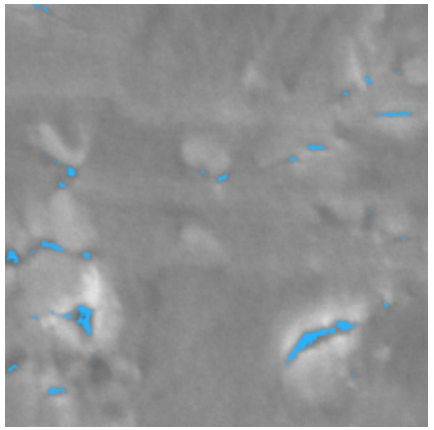
3. flatten_background

- What it does. Subtracts the blurred copy, so what is left is local contrast.
- What it costs. It is a high-pass filter, so it also erases anything wider than its own radius. The detector's own docstring measures this: pixels t...

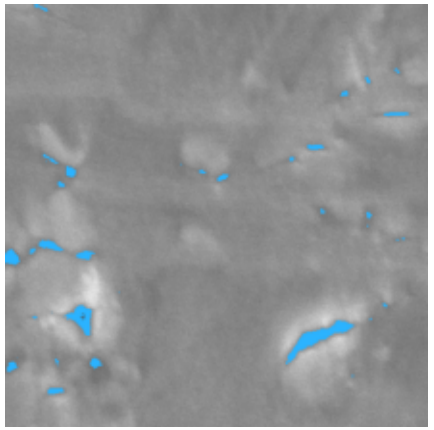


4. The darkness cut, at one radius

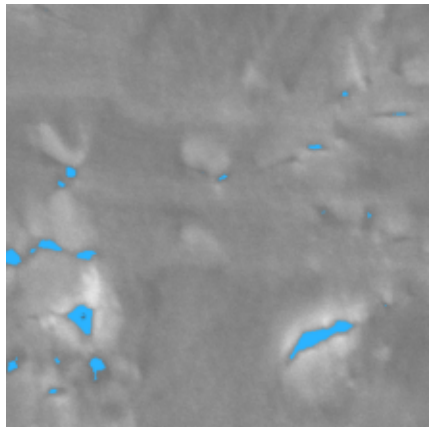
- What it does. Keeps a pixel if the denoised flat image is below min(Otsu, median − 5×MAD), OR if the raw 8-bit image is below 10. Taking the lower ...
- What it buys, on this crop. precision 0.9900, recall 0.2587, IoU 0.2580 against the hand paint. Almost everything it marks is crack; it finds about...



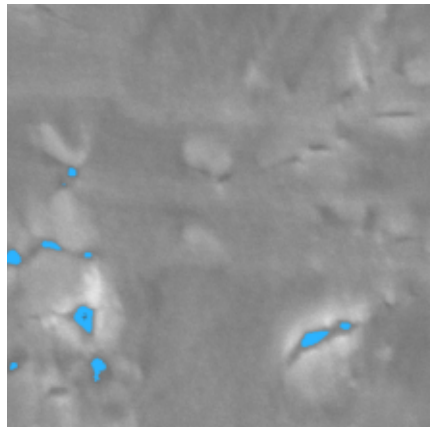
sigma = 5 · 2,155 px here · 2,155 of them new ·



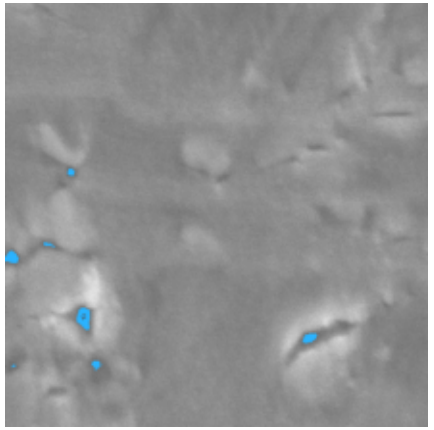
sigma = 10 · 3,456 px here · 1,382 of them new ·



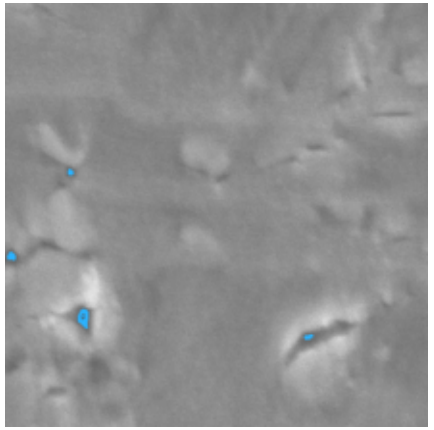
sigma = 20 · 3,012 px here · 292 of them new ·



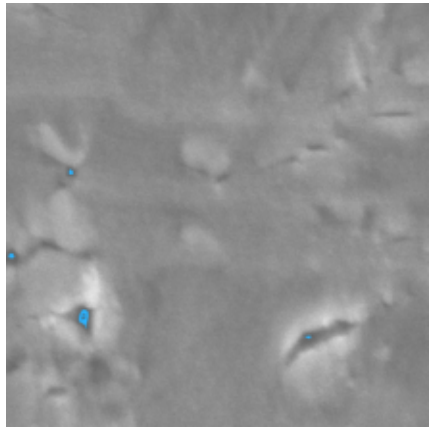
sigma = 40 · 1,793 px here · 45 of them new · 250,889



sigma = 80 · 823 px here · 0 of them new · 266,073 px



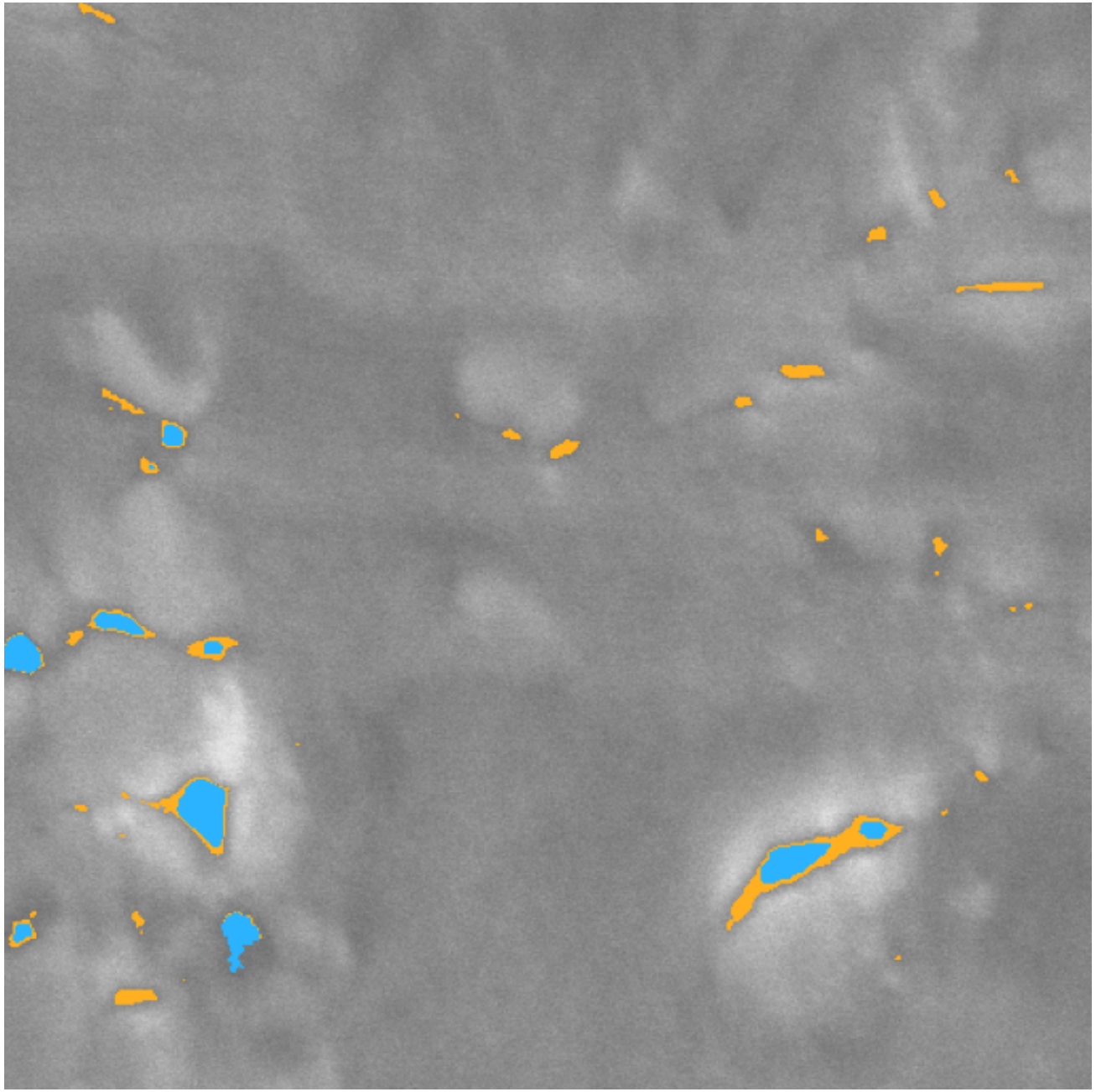
sigma = 160 · 400 px here · 0 of them new · 237,561



sigma = 320 · 230 px here · 0 of them new · 192,566

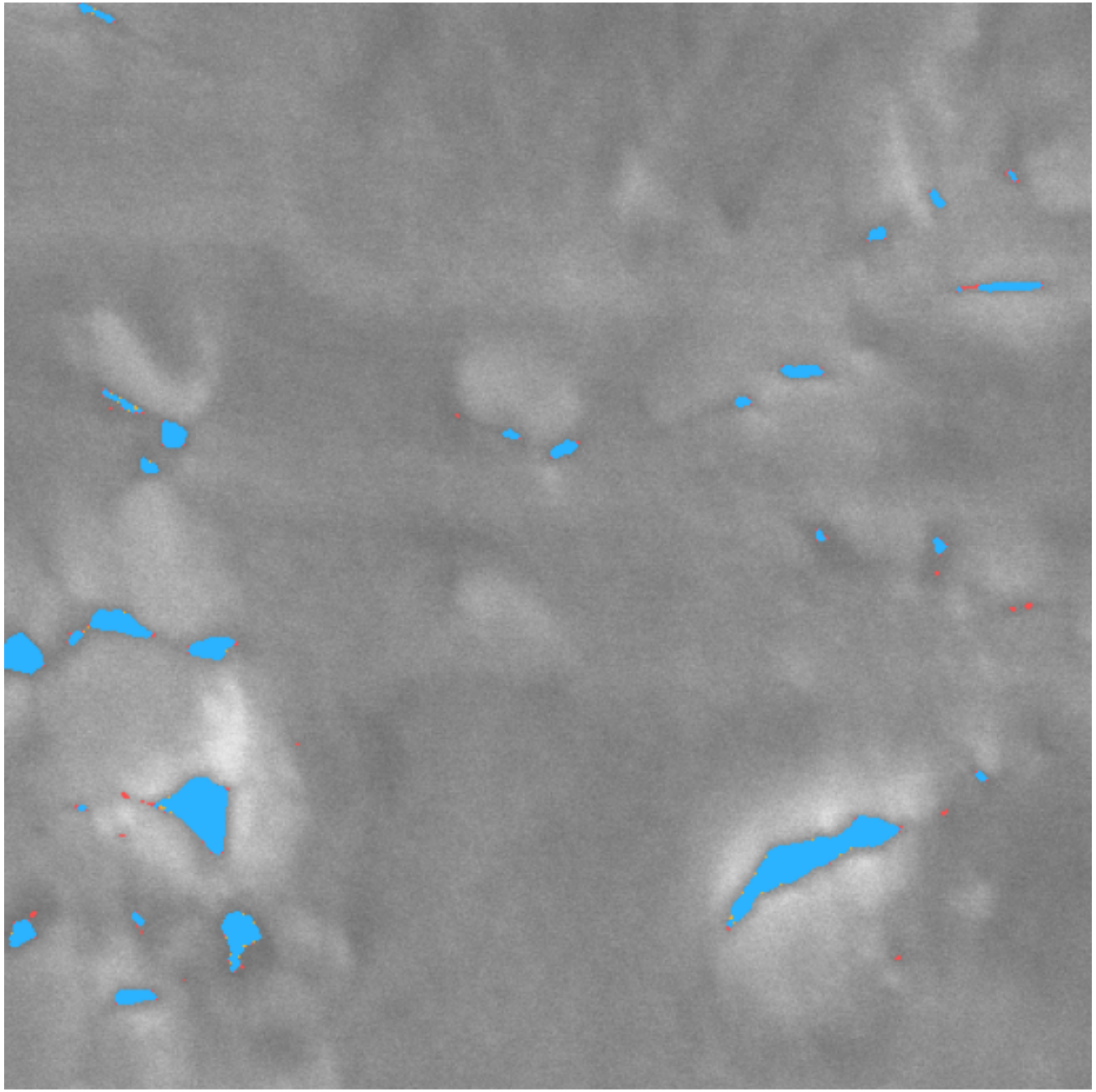
5. The same cut at each of the seven radii

- What it does. Re-flattens the image at sigma = 5, 10, 20, 40, 80, 160, 320 and takes the identical cut at each. A crack is only reliably dark at th...
- “New” counts pixels no earlier rung and not the absolute-darkness term had already claimed, taken in ascending sigma, so the seven new-counts sum t...



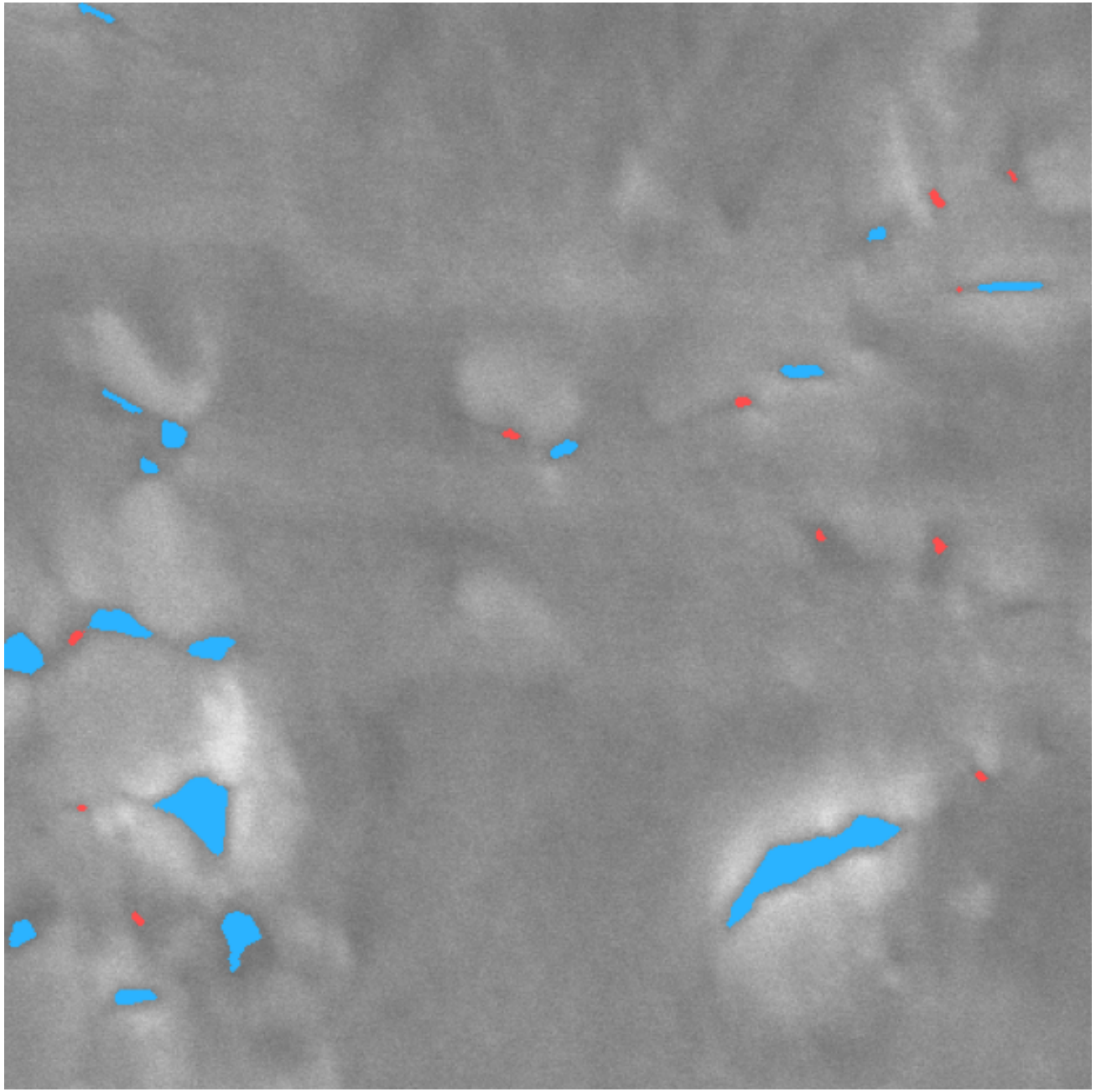
6. Their union — the change v1.49 made

- What it does. ORs the seven cuts. It is a strict superset of the single-radius mask by construction, so it cannot drop a pixel the older build kept...
- What it buys, on this crop. +2,081 px here and +183,944 px frame-wide. precision 0.9900 → 0.8138, recall 0.2587 → 0.4574, IoU 0.2580 → 0.4141. Over...



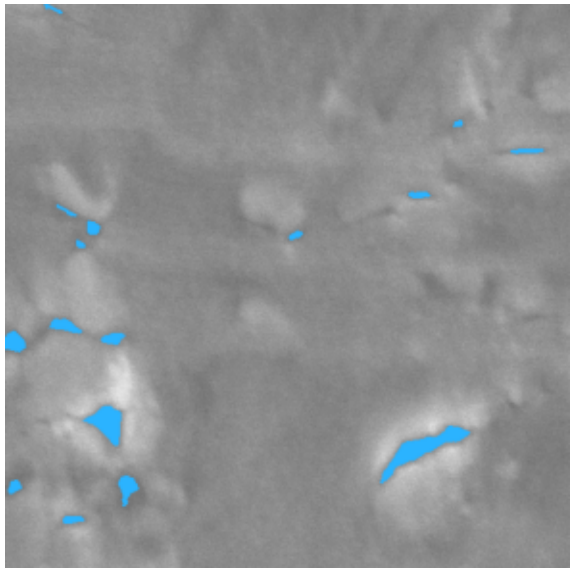
7. clean_mask

- What it does. Drops components of 13 px or fewer, opens with a disk of radius 1, closes with a disk of radius 3, fills holes.
- What it costs here. removes 168 px, adds back 42 (the closing and the hole fill are extensive), net -126 px in this crop.

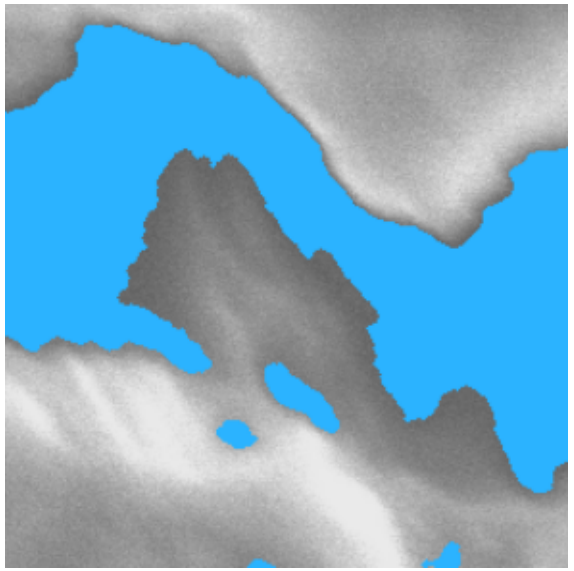


8. The 40 px region floor

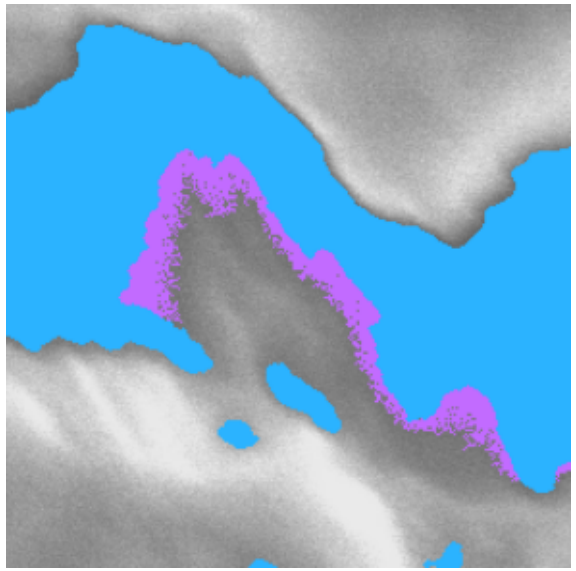
- What it does. extract_candidates labels the clean mask with connectivity=2 and keeps only components of at least 40 px, renumbering the survivors 1...
- What it costs here. 249 px in this crop.



carries 3,514 px in this crop (1.34% of it) · 704,310 px in the



A different 256 px window, at x=0 y=992: the mask before Pass 2.



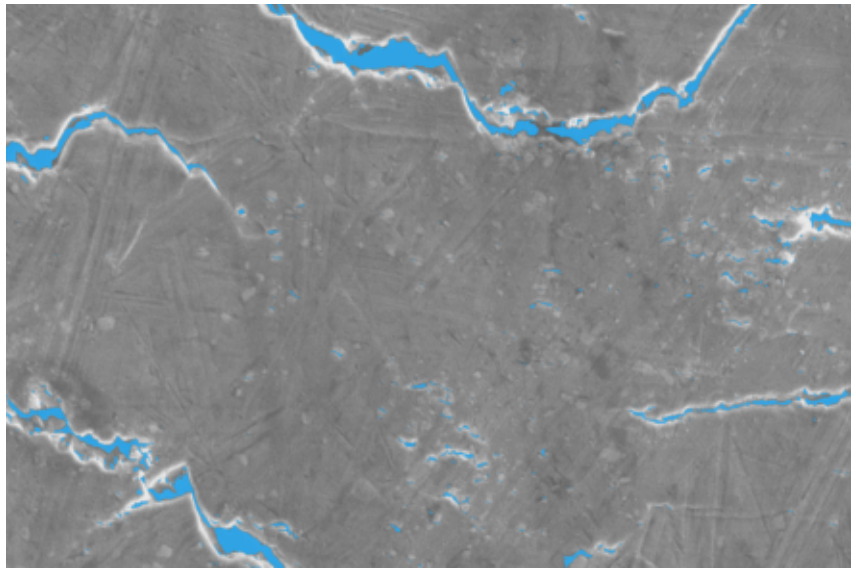
The same window after. Violet is Pass 2's addition — its largest

9. Pass 2

- What it does. With a crack mask now in hand, proposes concavity, bridge-corridor and interior-fill candidates relative to it, scores them with the ...
- What it buys. +41,369 px frame-wide (5.87% of the final mask) and 0 px in this crop. The second pair of panels is a different place, not a differen...



carries 3,514 px in this crop (1.34% of it) · 704,310 px in the whole frame



The same mask over the whole frame, for scale.

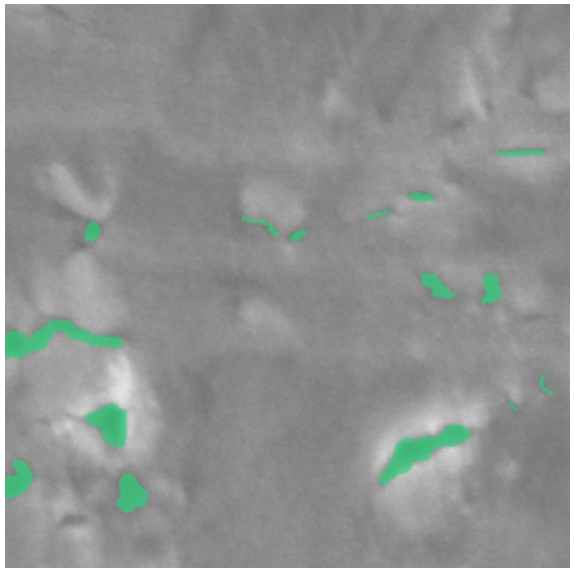
10. The emitted mask

- What it does. Unions every kept region into one boolean mask and saves it. Area fraction 2.799% on this frame, 1.341% in this crop.

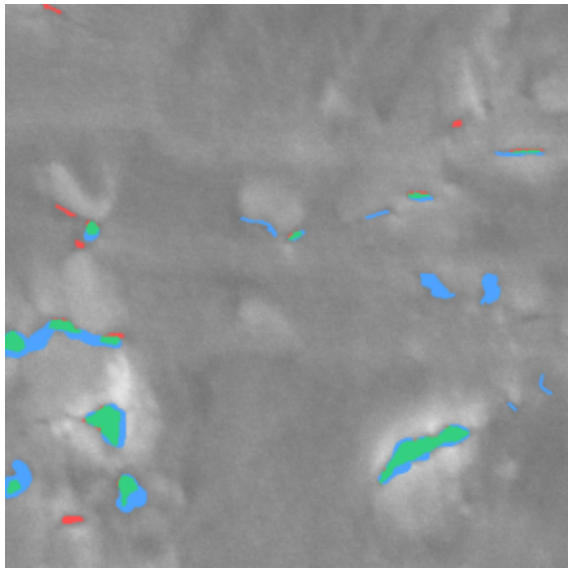


11. The measured output

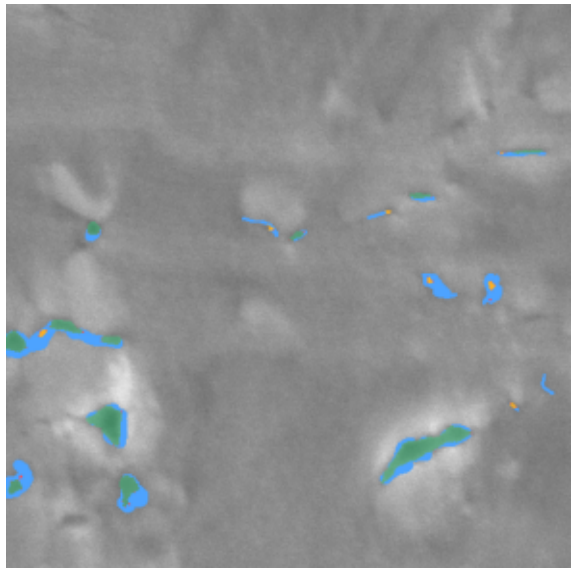
- What it does. Re-labels the emitted mask and measures every component: area, tip-to-tip geodesic, skeleton length, mean and max width, tortuosity, ...
- What it costs. 452 rows on this frame, 16 in this crop. Length is a bracket, not a value: 2.65% of regions on this frame touch the edge, but they a...



The hand paint for this crop: 6,919 px, 2.64% of it. Painted



green = found, red = marked but not painted, blue = painted but



The missed paint, charged to the stage that dropped it: blue =

12. Against the hand paint, and where the recall goes

- On this crop: precision 0.8557, recall 0.4346, IoU 0.4049. One crop is not a corpus figure.
- The two stages worth marking are the first row and everything else. On this crop the darkness cut owns 95.19% of all missed paint and the two morph...

⚠ How to read any number on this page. These are one crop of one frame, shown because the stages are comparable on it. The corpus figures, over the 26 hand-labelled tile...